Consider a polar curve $r = r(\theta)$. The slope of the tangent line to the curve at a point $(x,y) = (r\cos(\theta), r\sin(\theta))$ is given by $\frac{dy}{dx}$. By the Chain Rule,

$$\frac{dy}{d\theta} = \frac{dy}{dx}\frac{dx}{d\theta}.$$

This is just the usual formula for the slope of the tangent line to a parametric curve $(x,y) = (x(\theta), y(\theta))$ with parameter θ. If $\frac{dx}{d\theta} \neq 0$, then we can solve for $\frac{dy}{dx}$ to obtain

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}.$$

By the Product Rule,

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(r\sin(\theta)) = \frac{dr}{d\theta}\sin(\theta) + r\cos(\theta),$$
$$\frac{dx}{d\theta} = \frac{d}{d\theta}(r\cos(\theta)) = \frac{dr}{d\theta}\cos(\theta) - r\sin(\theta),$$

and therefore

$$\boxed{\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin(\theta) + r\cos(\theta)}{\frac{dr}{d\theta}\cos(\theta) - r\sin(\theta)}}. \tag{1}$$

Rather than memorizing Equation (1), however, it is often easier to just compute $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$ and divide them.

Example.

(a) Find the slope of the tangent line for the cardioid $r = 1 + \sin(\theta)$ when $\theta = \frac{\pi}{3}$.

(b) Find the points on the cardioid where the tangent line is horizontal or vertical.

Solution.

(a) We have

$$y = (1+\sin(\theta))\sin(\theta) = \sin(\theta) + \sin^2(\theta)$$
$$x = (1+\sin(\theta))\cos(\theta) = \cos(\theta) + \frac{1}{2}\sin(2\theta)$$

so

$$\frac{dy}{d\theta} = \cos(\theta) + 2\sin(\theta)\cos(\theta)$$
$$= \cos(\theta) + \sin(2\theta)$$

and

$$\frac{dx}{d\theta} = -\sin(\theta) + \cos(2\theta),$$

therefore

$$\frac{dy}{dx} = \frac{\cos(\theta)+\sin(2\theta)}{-\sin(\theta)+\cos(2\theta)}.$$

Evaluating at $\theta = \frac{\pi}{3}$ gives

$$\begin{aligned}\frac{dy}{dx}\Big|_{\theta=\frac{\pi}{3}} &= \frac{\cos(\frac{\pi}{3})+\sin(\frac{2\pi}{3})}{-\sin(\frac{\pi}{3})+\cos(\frac{2\pi}{3})} \\ &= \frac{\frac{1}{2}+\frac{\sqrt{3}}{2}}{-\frac{\sqrt{3}}{2}-\frac{1}{2}} \\ &= -\frac{\frac{1}{2}+\frac{\sqrt{3}}{2}}{\frac{1}{2}+\frac{\sqrt{3}}{2}} \\ &= \boxed{-1}.\end{aligned}$$

(b) The tangent line is horizontal when $\frac{dy}{dx} = 0 \iff \frac{dy}{d\theta} = 0, \frac{dx}{d\theta} \neq 0$ which implies

$$\begin{aligned}0 &= \cos(\theta)+\sin(2\theta) \\ &= \cos(\theta)+2\sin(\theta)\cos(\theta) \\ &= \cos(\theta)(1+2\sin(\theta))\end{aligned}$$

which implies either $\cos(\theta) = 0$ or $1+2\sin(\theta) = 0 \iff \sin(\theta) = -\frac{1}{2}$. We have $\cos(\theta) = 0$ when θ is a half integer multiple of $\frac{\pi}{2}$. Since $0 \leq \theta \leq 2\pi$, this means $\theta = \frac{\pi}{2}$ or $\frac{3\pi}{2}$. We have $\sin(\theta) = -\frac{1}{2}$. We have $\sin(\theta) = -\frac{1}{2}$ when $\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$ or $\theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$. Thus, the solutions are $\boxed{\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}}$.

The tangent line is vertical when $\frac{dx}{d\theta} = 0$, which implies

$$\begin{aligned}0 &= -\sin(\theta)+\cos(2\theta) \\ &= -\sin(\theta)+1-2\sin^2(\theta) \\ &= 2\sin^2(\theta)+\sin(\theta)-1 \\ &= (2\sin(\theta)-1)(\sin(\theta)+1)\end{aligned}$$

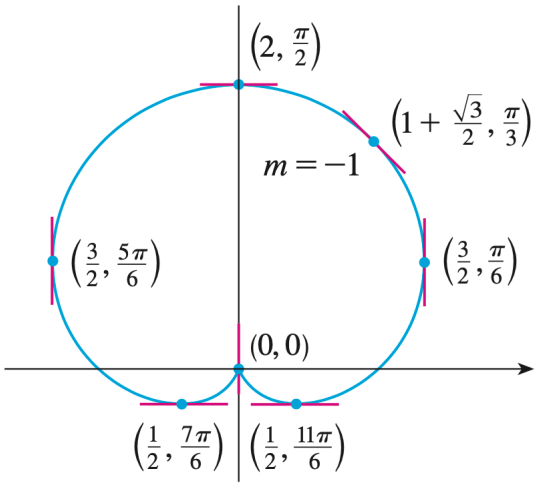
which implies $\sin(\theta) = \frac{1}{2}$ or $\sin(\theta) = -1$. The former occurs when $\theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$ and the latter occurs when $\theta = \frac{3\pi}{2}$. Therefore the solutions are $\boxed{\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}}$.

Plugging these values of θ into $r(\theta) = 1+\sin(\theta)$, the points on the cardioid where the tangent line is horizontal are $\boxed{\left(2, \frac{\pi}{2}\right), \left(\frac{1}{2}, \frac{7\pi}{6}\right), \left(\frac{1}{2}, \frac{11\pi}{6}\right)}$ and the points on the cardioid where the tangent line is vertical are $\boxed{\left(\frac{3}{2}, \frac{\pi}{6}\right), \left(\frac{3}{2}, \frac{5\pi}{6}\right)}$.

When $\theta = \frac{3\pi}{2}$, both $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$ are 0, so we must be careful.

$$\begin{aligned}
\lim_{\theta\to\frac{3\pi}{2}^-}\frac{dy}{dx} &= \lim_{\theta\to\frac{3\pi}{2}^-}\frac{\cos(\theta)+\sin(2\theta)}{-\sin(\theta)+\cos(2\theta)}\\
&= \lim_{\theta\to\frac{3\pi}{2}^-}\frac{\cos(\theta)+2\sin(\theta)\cos(\theta)}{-\sin(\theta)+1-2\sin^2(\theta)}\\
&= \lim_{\theta\to\frac{3\pi}{2}^-}\frac{\cos(\theta)(1+2\sin(\theta))}{(1+\sin(\theta))(1-2\sin(\theta))}\\
&= \lim_{\theta\to\frac{3\pi}{2}^-}\frac{\cos(\theta)}{(1+\sin(\theta))}\frac{(1+2\sin(\theta))}{(1-2\sin(\theta))}\\
&= \underbrace{\lim_{\theta\to\frac{3\pi}{2}^-}\frac{\cos(\theta)}{(1+\sin(\theta))}}_{\text{indeterminate}}\underbrace{\lim_{\theta\to\frac{3\pi}{2}^-}\frac{(1+2\sin(\theta))}{(1-2\sin(\theta))}}_{=-1/3}\\
&= -\frac{1}{3}\lim_{\theta\to\frac{3\pi}{2}^-}\frac{\cos(\theta)}{(1+\sin(\theta))}\\
&= -\frac{1}{3}\lim_{\theta\to\frac{3\pi}{2}^-}\frac{-\sin(\theta)}{\cos(\theta)} \quad \text{(by l'Hopital's rule)}\\
&= \frac{1}{3}\lim_{\theta\to\frac{3\pi}{2}^-}\tan(\theta)\\
&= \infty
\end{aligned}$$

and $\lim_{\theta\to\frac{3\pi}{2}^+}\frac{dy}{dx} = \frac{1}{3}\lim_{\theta\to\frac{3\pi}{2}^+}\tan(\theta) = -\infty$ so there is a vertical tangent at the pole.



Exercises

1.) Find the slope of the tangent line to the give polar curve at the point specified by the value of θ.

(a) $r = 2\sin(\theta)$, $\theta = \frac{\pi}{6}$.

(b) $r = \frac{1}{\theta}$, $\theta = \pi$.

(c) $r = \cos(2\theta)$, $\theta = \frac{\pi}{4}$

2.) Find the points on the give curve where the tangent line is horizontal or vertical.

(a) $r = 3\cos(\theta)$

(b) $r = 1 + \cos(\theta)$

Answers: 1.) (a) $\sqrt{3}$, (b) $-\pi$, (c) 1; 2.) (a) Horizontal at $(3/\sqrt{2}, \pi/4), (-3/\sqrt{2}, 3\pi/4)$; vertical at $(3,0), (0,\pi/2)$; (b) Horizontal at $(\frac{3}{2}, \pi/3), (0,\pi)$ (the pole), and $(\frac{3}{2}, 5\pi/3)$; vertical at $(2,0), (\frac{1}{2}, 2\pi/3), (\frac{1}{2}, 4\pi/3)$.